

Generation Interconnection System Impact Study Report

For Queue Project Q24-2475

Mazomanie Industrial 138 kV
December 29, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2475. The Interconnection Customer proposes a 150.0 MW solar photovoltaic facility interconnecting at the Mazomanie Industrial 138 kV substation in the Pennsylvania area. The request was received on November 15, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 414 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,381,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The WNP NO. 2 CONSTRUCTION – LAKIN MUNICIPAL 138 kV Line is overloaded to 102.6% of its 280 MVA emergency rating for the loss of the LAKIN MUNICIPAL SUBSTATION 345/138 kV Auto. The project contributes 6.0% DFAX (9.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WNP NO. 2 CONSTRUCTION – LAKIN MUNICIPAL 138 kV Line	Reconductor line	\$1,381,000	100% (\$1,381,000)
	Total allocated to project		\$1,381,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.